

COMP3018

Human-Robot Interaction

20 CREDIT MODULE

ASSESSMENT: 100% Coursework **W1: 30% Report Outline**
W2: 70% Report

MODULE LEADER: Dr. Amir ALY

MODULE AIMS

- Equip students with an understanding of multimodal human-robot interaction and its applications in daily life.
- Enable students to create, and evaluate, a robot system for interaction with users.
- Provide insights into the current and future challenges facing human-robot interaction.

ASSESSED LEARNING OUTCOMES (ALO):

1. Explain the elements included in the field of human-robot interaction.
2. Describe what differentiates cognitive robotics from the other sub-branches of robotics.
3. Design a human-robot interaction experiment and evaluate the performance of the system through appropriate metrics and evaluation methods.

Overview

This document contains all the necessary information pertaining to the assessment of *COMP3018 Human-Robot Interaction*. The module is assessed via **100% coursework**, across two elements: *30% Set Exercises* and *70% Report*.

The sections that follow will detail the assessment tasks that are to be undertaken. The submission and expected feedback dates are presented in Table 1. All assessments are to be submitted electronically via the respective DLE module pages before the stated deadlines.

	Submission Deadline	Feedback
Set Exercise (30%)	12th March 2026 (15:00)	31st March 2026 (15:00)
Report (70%)	5th May 2026 (15:00)	30th May 2026 (15:00)

Table 1: Assessment Deadlines

All assessments will be introduced in class to provide further clarity over what is expected and how you can access support and formative feedback prior to submission. **Whilst the assessment information is provided at the start of the module, it is not necessarily expected that you will start this immediately**, as you will often not have sufficient understanding of the topic. The module leader will provide guidance in this respect.

Moderation

This assignment brief has been moderated in line with the university policy.

Moderator Name	Moderation Date
Shaymaa Al-Juboori	09/02/2026

Useful Assignment Information

Please see below some useful information regarding submissions for your modules.

Good Coding Practices

Like all things, no code is perfect. Just because your code compiles and runs does not mean it is perfect, there is always room for improvement. No code will achieve 100% of the available marks. Where code submission is required by a module, it will be assessed against the following criteria:

1) Functionality

- a. Does your code meet all the specified requirements of the assignment?
- b. Does your code behave correctly across typical and edge-case scenarios?
- c. Does your code have appropriate error handling that does not lead to it crashing or undefined behaviour?

2) Efficiency

- a. Is your code optimised in terms of performance and resource use?
- b. Does your code handle functions, variables, data calls efficiently?
- c. Is there any unnecessary repetition, complexity or processing reducing efficiency within your code?

3) Readability

- a. Is your code logically structured and easy to read?
- b. Have you maintained standard formatting conventions like indentations, spacing and naming, within your code?
- c. Are variables, functions, and classes clearly named and purposeful?

4) Documentation

- a. Are there helpful comments explaining non-obvious parts of your code?
- b. Do your comments document your process, development or rationale clearly?
- c. Could someone unfamiliar with the code understand the approach you have taken?

Use of Generative AI for Creating Code

Each assignment element for each module will have a clear indication of the permitted level of use of AI (solo, assisted or partnered) including the generation of code. Please ensure that you **read and understand the permitted use**. If you are unsure of a particular use, please reach out to the Module Leader and ask.

We strongly encourage you to read and explore beyond the core content delivered within the modules. However, this must be done correctly following academic process. Wherever code is drawn from an outside source (e.g. you have not written it yourself), regardless of whether this is AI generated or from an online repository (GitHub, Stack Overflow etc) **you must reference the original source**. This can be done in your documentation as comments or part of your write ups. Students found to be utilising code without referencing could face an academic offence. **If in doubt speak to your Module Leader.**

Versioning

A critical part of a development (software or academic) is versioning. Keeping a number of iterations over the course of development ensures you always have backups to fall back on. It also provides clear demonstration of the development process you implemented.

Using online/cloud-based storage solutions and repositories provide additional peace of mind, alongside being industry practice. The university provide OneDrive to all students which offers inbuilt version history for Microsoft products. GitHub is the university recommended repository for versioning code, which includes great integration with a number of IDEs.

Submitting Code

It is vital that you confirm that all code that you submit is in the **correct format** and **compiles correctly** or **runs without error**. If you clean up your code before submission, please ensure that all dependencies (libraries, functions and variables) are included so that the marker can compile your code.

We can only mark what has been submitted. If the code does not run correctly, it is not our responsibility to spend time error handling to award you marks.

It is also important to **show your working**. Tidying up your code is a critical part of coding practices, but if you remove the workings that provides you with the required output, we cannot see how you got there.

In a worst-case scenario, if you remove a key part of your working, and on compiling your code it gives an error or different result, we have no way of confirming what went wrong, or indeed if you fabricated those outputs.

Please note, just because a piece of code has given the wrong output, it does not mean it should be deleted. Seeing these attempts with comments documenting your attempt allows us to understand where you may have made an error and provide feedback that addresses this. In some cases, you could also be awarded marks for the attempt.

Assessment 1: Set Exercise

Acceptable level of generative AI tool use in this assessment element

The acceptable level of GenAI use for this element is detailed with the allowed uses listed below. This is split into three categories (**Solo Work** – work must be your own with no AI support, **Assisted Work** – some uses of AI tools allowed, **Partnered Work** – AI tools integral part of the work). If you have any questions, please contact the module leader.

Solo Work	You must not use generative AI tools.	☐
Assisted Work	You are permitted to use generative AI tools in an assistive role.	☒
Partnered Work	Generative AI tool use is required as an integral part of the assessment, but transparency is required.	☐

Table 2: Acceptable Level of generative AI use

Inline with the indicated acceptable level of AI use above, the following uses are acceptable:

- **A2 - Planning & Structuring Projects**
- **A4 – Research Assistance**

Any use of AI in your work must be declared within your documentation. You **must also include a signed Generative AI Declaration as an appendix to your submission**. The declaration form can be found on DLE (or on the Programme Page). This form will not be included in any word count associated with this assignment.

Task (1) Cultural Differences and HRI Design (40% of the total mark of assessment 1)

Consider the following instructions and discuss the following points in a report in detail. You should use relevant papers in the literature to answer these questions.

1. Summarize the cultural factors discussed in Kaplan (2004) that impact the acceptance of humanoid robots in the West and the East.
2. Based on your knowledge and experience, propose cultural factors that might have an impact on the acceptance of humanoid robots in Africa.
3. Suggest ways in which these cultural factors can be used to determine traits (e.g., in appearance or behavior) that would make humanoid robots more acceptable in (a) the East, (b) the West, and (c) Africa.
4. Suggest how these traits can be used to adapt or refine the design patterns for sociality in human-robot interaction proposed by Kahn et al. (2008) for (a) the East, (b) the West, and (c) Africa.

References

- 1- Kaplan, F. (2004) "Who is afraid of the humanoid? Investigating cultural differences in the acceptance of robots". In *International Journal of Humanoid Robotics*, 1(3): 1–16.

- 2- Kahn, P. H., et al. (2008) 'Design patterns for sociality in human-robot interaction'. In *Proceedings of the 3rd ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, pages 97–104, Amsterdam, Netherlands.

Submission Requirements

- Justify your answers in detail through necessary discussion and analysis.
- Any references should be appropriately cited in the report using the Harvard referencing style.
- Report submitted as a PDF document. The number of words should not exceed 1,750.
- Appendix with completed Generative AI Declaration.

Assessment Criteria

The following marking scheme will be used for this part of the assessment:

- **Part (1- 1)** (20% of the task mark)
- **Part (1- 2)** (20% of the task mark)
- **Part (1- 3)** (30% of the task mark)
- **Part (1- 4)** (30% of the task mark)

Task (2) Models for HRI (60% of the total mark of assessment 1)

The Partially-Observable Markov Decision Process (POMDP) is one of the widely used models for human-robot collaboration. Discuss the following points in detail (You should use relevant papers in the literature to support your discussion and analysis):

1. Discuss the efficient role that POMDP plays in modelling and solving problems related to trust, cooperation, coordination, and collaboration in human-robot teams.
2. Analyze the role of uncertainty in human-robot collaboration and how POMDPs can be used to make decisions in uncertain environments.
3. Discuss the challenges of modelling trust in human-robot collaboration and how POMDPs can help address them.
4. Develop a POMDP model for a specific scenario in human-robot collaboration and explain how it takes into account trust and uncertainty. What are the benefits and limitations?
5. Discuss the ethical and social implications of using POMDPs in human-robot collaboration.

Submission Requirements

- Justify your answers in detail through necessary discussion and analysis.
- Any references should be appropriately cited in the report using the Harvard referencing style.
- Report submitted as a PDF document. The number of words should not exceed 1,650.
- Appendix with completed Generative AI Declaration.

Assessment Criteria

The following marking scheme will be used for this part of the assessment:

- **Part (2- 1)** (20% of the task mark)
- **Part (2- 2)** (20% of the task mark)
- **Part (2- 3)** (15% of the task mark)
- **Part (2- 4)** (35% of the task mark)
- **Part (2- 5)** (10% of the task mark)

Threshold Criteria (indicative)

< 40% Little or no analysis, and answers are largely incorrect. Little understanding of the subject. Almost no evidence of investigation and research for answering the questions. The writings are neither clear nor well-written or structured.

40–49% Brief discussion and little analysis for the different tasks. Answers are partially correct and/or complete, and missing elaborate details and explanations. There is little evidence of investigation and research for answering the questions. The content and structure of the writings are appropriate to a limited extent.

50–59% The discussion and analysis for the different tasks are relatively detailed and clear. Different answers are correct and complete with a sufficient level of detail explaining how the results are obtained. There is medium evidence of investigation and research for answering the questions. The writings are of relatively good standards and quality.

60–69% Detailed discussion and analysis for the different tasks. A significant majority of answers are correct and complete, with a very good level of detail explaining clearly how results are obtained. There is very good evidence of investigation and research for answering the questions. The writings are of very good standards and quality.

≥ 70% The different tasks are very well discussed in detail, supported by excellent arguments. Answers are correct and complete, especially with clear and well-justified analysis and description. There is strong evidence of investigation and research for answering the questions (e.g., deep analysis and full investigation). The writings are of high standards and quality (focused and concise).

Assessment 2: Report

Acceptable level of generative AI tool use in this assessment element

The acceptable level of GenAI use for this element is detailed with the allowed uses listed below. This is split into three categories (**Solo Work** – work must be your own with no AI support, **Assisted Work** – some uses of AI tools allowed, **Partnered Work** – AI tools integral part of the work). If you have any questions, please contact the module leader.

Solo Work	You must not use generative AI tools.	☐
Assisted Work	You are permitted to use generative AI tools in an assistive role.	☒
Partnered Work	Generative AI tool use is required as an integral part of the assessment, but transparency is required.	☐

Table 4: Acceptable Level of generative AI use

Inline with the indicated acceptable level of AI use above, the following uses are acceptable:

- **A2 - Planning & Structuring Projects**
- **A4 – Research Assistance**

Any use of AI in your work must be declared within your documentation. You **must also include a signed Generative AI Declaration as an appendix to your submission**. The declaration form can be found on DLE (or on the Programme Page). This form will not be included in any word count associated with this assignment.

Task (3) Literature Review (40% of the total mark of assessment 2)

You have studied different topics in human-robot interaction. Write an essay about **assistive robotics and its applications**, describing its impact, the challenges it faces, future applications, and ethical issues that need to be considered

Submission Requirements

- Any references should be appropriately cited in the report using the Harvard referencing style.
- The references that you will use should be peer-reviewed journal papers or conference papers.
- Report submitted as a single PDF document. The essay should not exceed 1,800 words.
- Appendix with completed Generative AI Declaration.

Assessment Criteria

The following marking scheme will be used for this part of the assessment:

- **Introduction (10% of the task mark)**
- **Literature review (20% of the task mark)**
- **Applications (30% of the task mark)**
- **Discussion (20% of the task mark)**
- **Conclusion (10% of the task mark)**
- **References (10% of the task mark)**

Task (4) Programming Project (60% of the total mark of assessment 2)

During this module, you've explored various aspects of human-robot interaction through lectures and workshops. Your assignment now is to conceptualize and suggest a unique project. This will provide you with an opportunity to engage more extensively with the intricacies of human-robot interaction.

- **The requirements for the project are as follows:**

1. The project should include a novel intellectual contribution that addresses a problem of interest, ideally demonstrating the application of this new idea through the use of robots or other empirical methods.
2. The focus of the chosen project should be, whether similar or related, to a core topic(s) we have covered in class.

The project could be about (but is not limited to) creating an exciting robot application (**after approving the idea with the module leader**), extending a model/application in a novel way, or designing a new approach to attack a problem relevant to the class

Submission Requirements

- Justify your answers with detailed explanations to demonstrate your level of understanding of the different parts of the task.
- Any references should be appropriately cited in the report using the Harvard referencing style.
- Submit **your code** and a **video of 5 minutes** describing your contribution and showing clearly a running demo of your work.
- Report submitted as a single PDF document. The number of words should not exceed 2,000.
- Appendix with completed Generative AI Declaration.

Assessment Criteria

The following marking scheme will be used for this part of the assessment:

- **Introduction (10% of the task mark)**
- **Background (10% of the task mark)**
- **Method and setup (35% of the task mark)**
- **Results/outcome/or system analysis (30% of the task mark)**
- **Conclusion (10% of the task mark)**
- **References (5% of the task mark)**

Threshold Criteria (indicative)

< 40% Little or no analysis, and answers are largely incorrect. Little understanding of the subject. Almost no evidence of investigation and research for answering the questions. The writings are not clear and not well-written or structured.

40–49% Brief discussion and little analysis for the different tasks. Answers are partially correct and/or complete, and missing elaborate details and explanations. There is little evidence of investigation and research for answering the questions. The content and structure of the writings are appropriate to a limited extent.

50–59% The discussion and analysis for the different tasks are relatively detailed and clear. Different answers are correct and complete with a sufficient level of detail explaining how the results are obtained. There is medium evidence of investigation and research for answering the questions. The writings are of relatively good standards and quality.

60–69% Detailed discussion and analysis for the different tasks. A significant majority of answers are correct and complete, with a very good level of detail explaining clearly how results are obtained. There is very good evidence of investigation and research for answering the questions. The writings are of very good standards and quality.

≥ 70% The different tasks are very well discussed in detail, supported by excellent arguments. Answers are correct and complete, especially with clear and well-justified analysis and description. There is strong evidence of investigation and research for answering the questions (e.g., deep analysis and full investigation). The writings are of high standards and quality (focused and concise).

Acceptable levels of AI use:

The table below provides the acceptable use categories for GenAI. Each assessment element may allow different uses. Please check the brief for each element carefully to see what uses are allowed.

Solo Work	S1 - Generative AI tools have not been used for this assessment.
Assisted Work	A1 – Idea Generation and Problem Exploration Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions). The interpretation and integration of research into the assignment remain the student's responsibility.
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.
	A11 - Other uses not listed above Please specify:
Partnered Work	P1 - Generative AI tool usage has been used integrally for this assessment Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify:

General Guidance

Extenuating Circumstances

There may be a time during this module where you experience a serious situation which has a significant impact on your ability to complete the assessments. The definition of these can be found in the University Policy on Extenuating Circumstances here:

<https://www.plymouth.ac.uk/student-life/your-studies/essential-information/exams/exam-rules-and-regulations/extenuating-circumstances>

Plagiarism

All of your work must be of your own words. You must use references for your sources, however you acquire them. Where you wish to use quotations, these must be a very minor part of your overall work.

To copy another person's work is viewed as plagiarism and is not allowed. Any issues of plagiarism and any form of academic dishonesty are treated very seriously. All your work must be your own and other sources must be identified as being theirs, not yours. The copying of another persons' work could result in a penalty being invoked.

Further information on plagiarism policy can be found here:

Plagiarism: <https://www.plymouth.ac.uk/student-life/your-studies/essential-information/regulations/plagiarism>

Examination Offences: <https://www.plymouth.ac.uk/student-life/your-studies/essential-information/exams/exam-rules-and-regulations/examination-offences>

Turnitin (<http://www.turnitinuk.com/>) is an Internet-based 'originality checking tool' which allows documents to be compared with content on the Internet, in journals and in an archive of previously submitted works. It can help to detect unintentional or deliberate plagiarism.

It is a formative tool that makes it easy for students to review their citations and referencing as an aid to learning good academic practice. Turnitin produces an 'originality report' to help guide you. To learn more about Turnitin go to:

<https://help.turnitin.com/new-links.htm?Highlight=guide>

The university also had Draft Coach available on the online version of Microsoft Word. Draft Coach is a Turnitin plugin and provides instant feedback on how to address citation issues, grammar mistakes and matches with Turnitin's database.

Referencing

The University of Plymouth Library has produced an online support referencing guide which is available here: <http://plymouth.libguides.com/referencing>.

Another recommended referencing resource is [Cite Them Right Online](#); this is an online resource which provides you with specific guidance about how to reference lots of different types of materials.